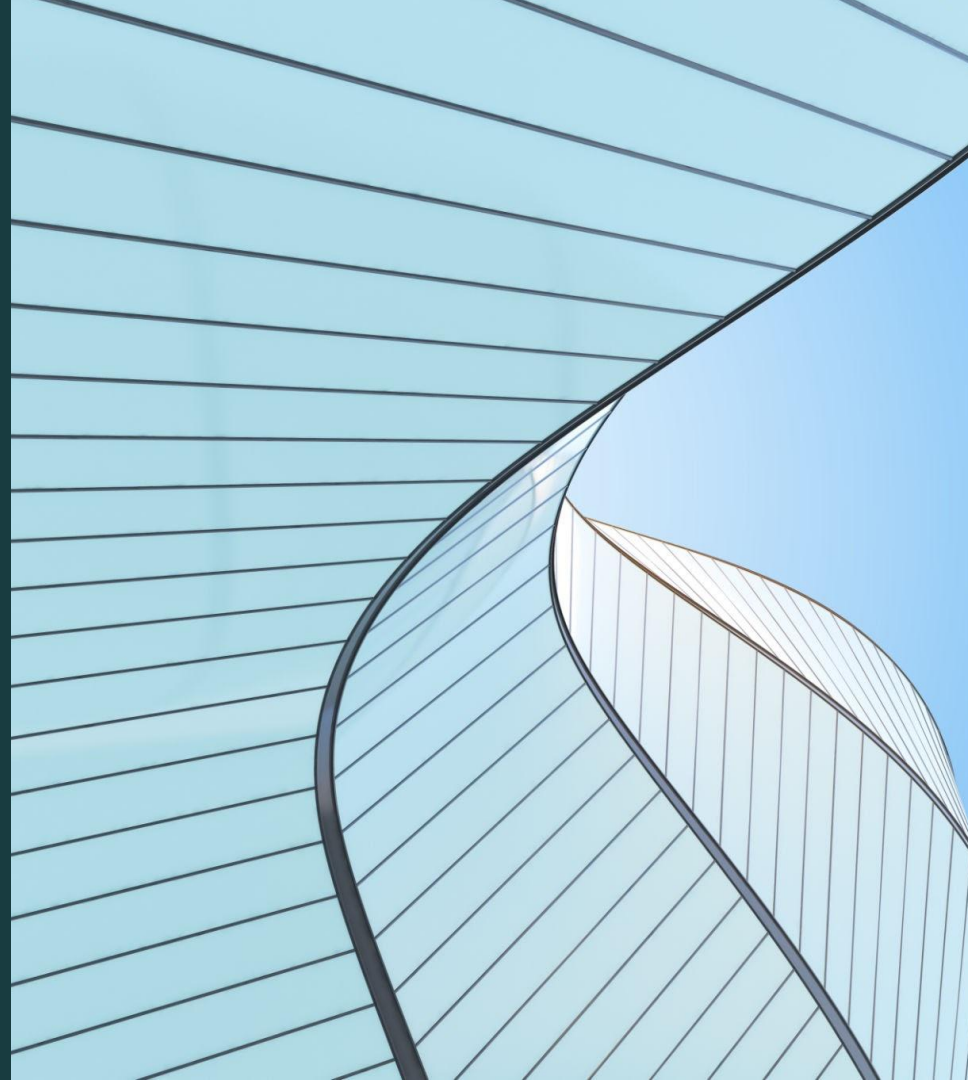


CIT304- Business Intelligence and Analytics

Prasan Ratnayake

Data Engineer | Superloop Tele Communications Pvt Ltd

0766799023

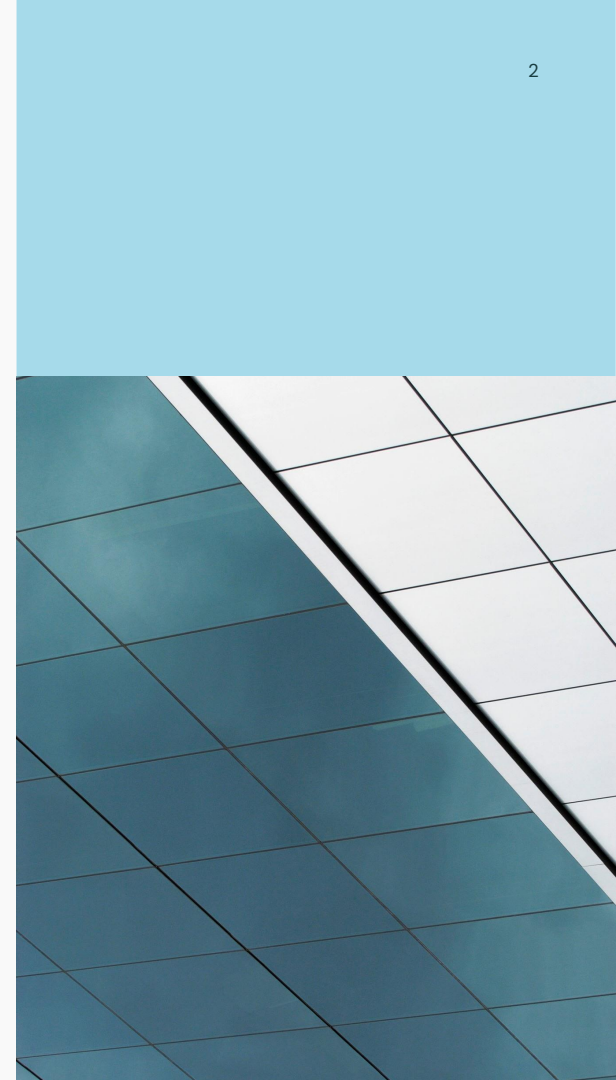


Course Outline

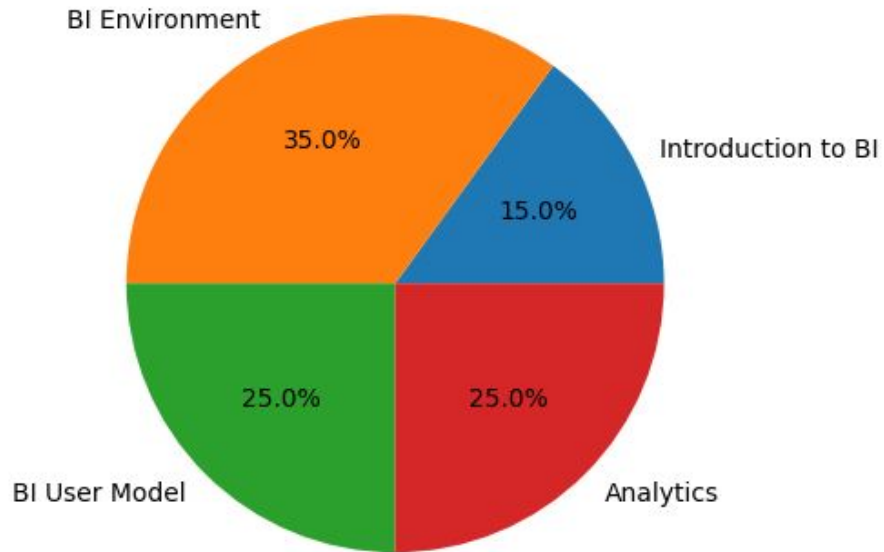
Outline Syllabus

1.	Introduction to Business Intelligence: What is BI?, History and Evolution, Business Intelligence Segments, Information and Intelligence, - Enterprise Business Intelligence, Business Intelligence Value Chain, Real time Business Intelligence, Business Intelligence applications.
2.	Business Intelligence Environment: Business models, Data models, Data sources, ETL tools, Data warehouse, Data marts, OLAP analysis and reporting tools. Types of Business Intelligence, Business Intelligence Platform - Applications in Business. Analytics -Dynamic roles in Business Intelligence - Challenges - Business Intelligence Tools – SAP, MicroStrategy, Zoho Analytics etc.
3.	Business Intelligence User Model: Opportunity Analysis, Content Management, End User Segmentation, Reporting and Querying, Online Analytical Processing Techniques (OLAP),Benefits of using OLAP, Dashboard, Key Performance Indicators OLAP Applications, report viewing, Analytical calculations, Predictive what-if.
4.	Analytics: Foundations and Technologies for decision making, Descriptive Analytics – Data warehousing, Visual Analysis, Business Performance. Predictive Analytics – Data Mining, Text Analysis, Text Mining, Sentiment Analysis, Web Analysis, Web Mining, Social Analytics

What is Business Intelligence?



BI Syllabus Distribution

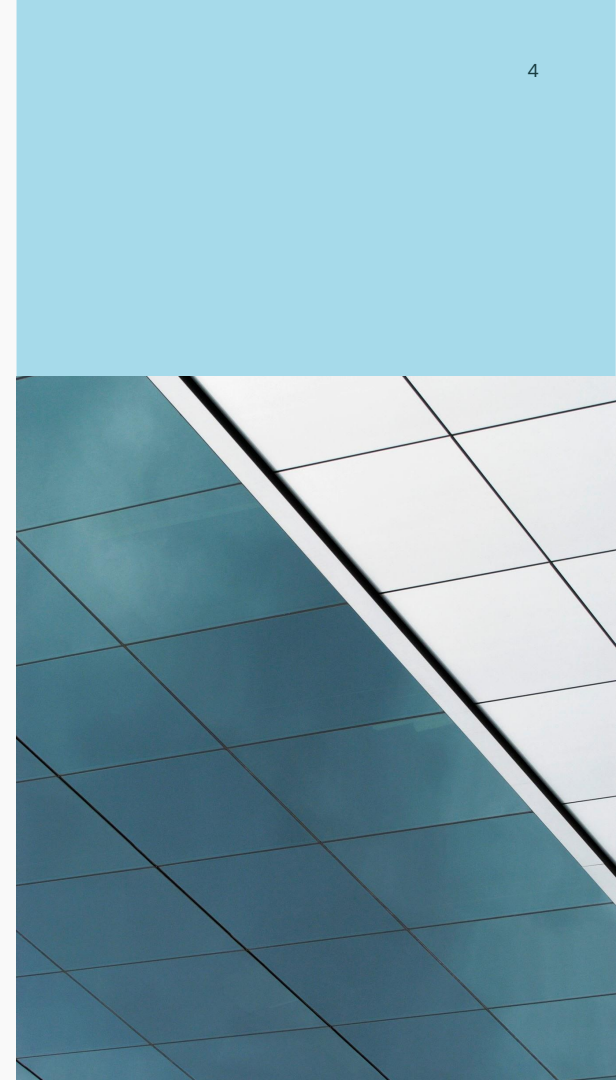


What is Business Intelligence?

Weekly Plan??

Week	Topics Covered
1	Introduction to BI, Evolution, BI vs Analytics, Business Value
2	BI in Business, Value Chain, BI Use Cases
3	Business Models & Domains, Stakeholder Data Needs
4	Data Sources, Data Requirements, Data Quality
5	Data Modeling Concepts, ER, Dimensional Models
6	Data Warehouse Concepts, Architectures
7	ETL Processes, Tools, Data Integration
8	DW Design, Data Marts, Data Cubes
9	OLAP, Reporting, Querying Tools
10	BI Platforms & Challenges
11	Power BI – Data Import, Modeling, Dashboards
12	Power BI – KPIs, Reports, Decision Analytics
13	Advanced Analytics, Social, Web, Text Analytics

What is Business Intelligence?



Assignments??

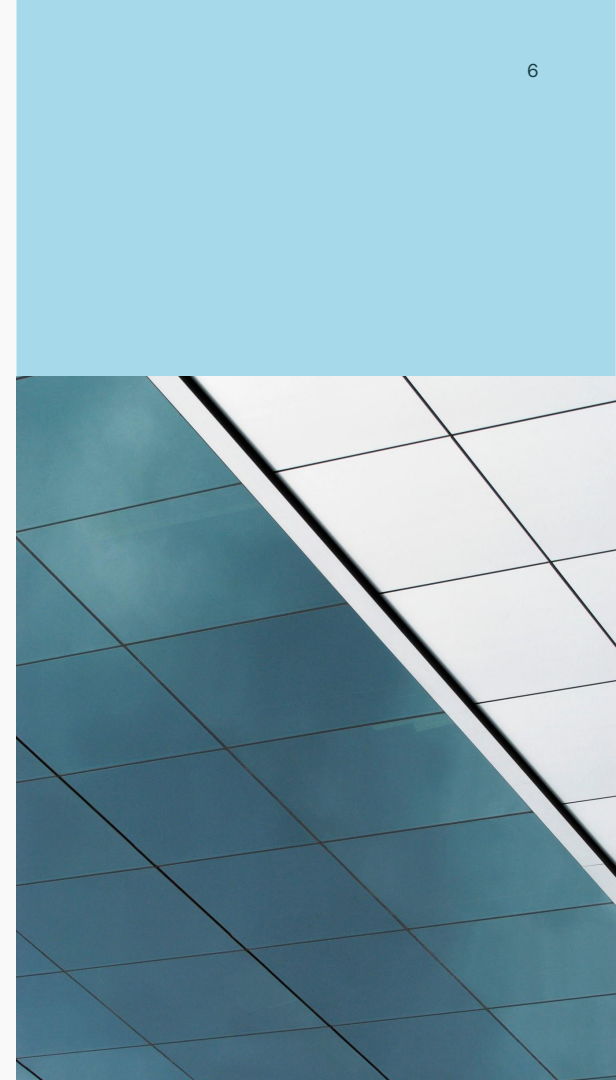
- 1. Mid-Semester Exam Paper (20%) - 20 mcqs*
- 2. Assignment I - Case-Study (10%)*
(5 separate case studies will be given)
- 3. Tableau / PowerBI Project (10%)*
- 4. Final Exam Paper (60%)*

What is Business Intelligence?

“...BI systems that combine data gathering, data storage and knowledge management with analysis to evaluate complex corporate and competitive information ... for improving the timeliness and quality of input to the decision process.”

– Solomon Negash & Paul Gray

What is Business Intelligence?



Global Management Systems

- **E-commerce Expansion**

- E-commerce worldwide expands to nearly \$3.6 trillion in 2019
- Growth in social commerce spurred by growth of mobile platform
- Mobile retail e-commerce growing more than 20 percent a year, reaching almost \$300 billion in 2020

- **Management Changes**

- Managers becoming more mobile
- Managers use social networks, collaboration tools
- Business intelligence applications accelerate

What is Business Intelligence?



Evolution of BI Systems

Decision Support Systems (DSS)

DSS are information systems that help managers make decisions, especially for semi-structured problems.

They combine **data access, analysis, and models** to support decision processes.

"DSS ... help decision makers compile useful information ... to identify and solve problems and make decisions."



Executive Information Systems (EIS)

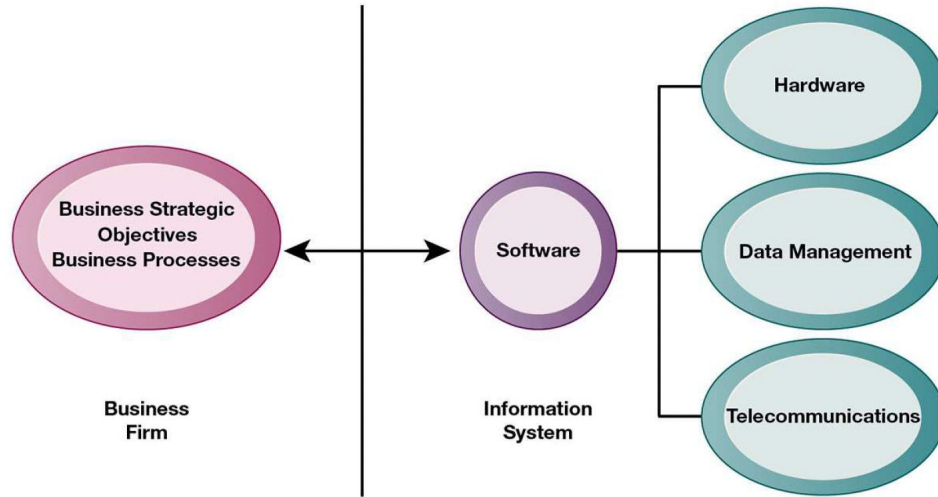
EIS emerged next to give **senior executives summary dashboards** and reports to make strategic decisions.



Business Intelligence (BI)

The term was popularized by **Howard Dresner** in 1989 to describe methods that use data for better decision-making.

The Interdependence Between Organizations and Information Systems



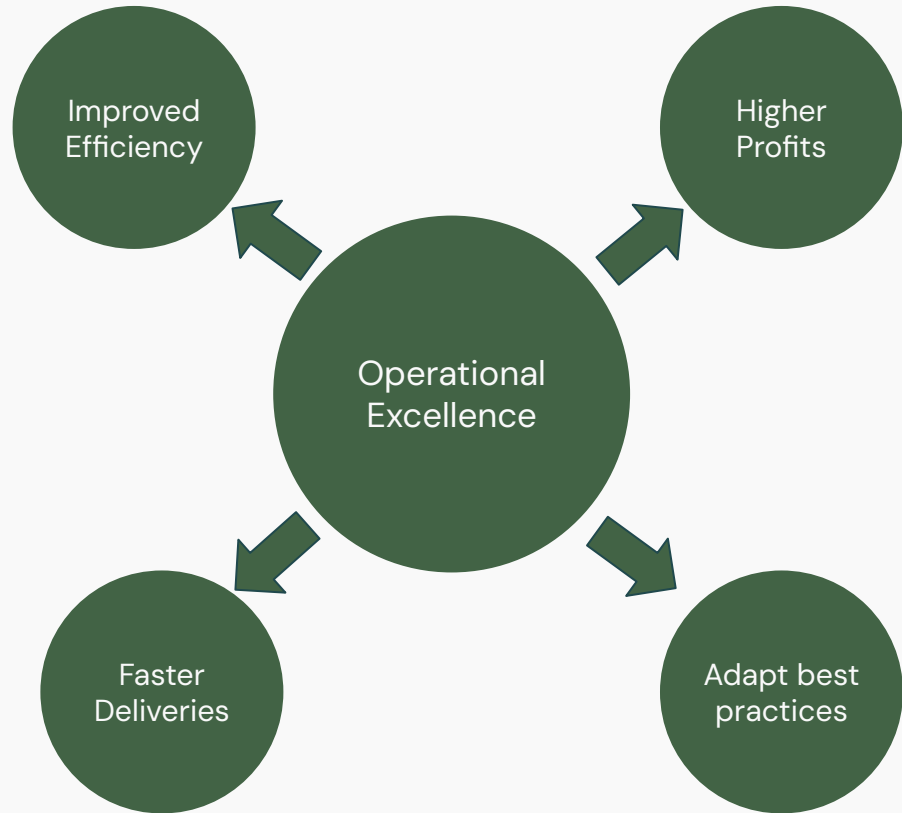
What is Business Intelligence?

Firms invest heavily in information systems to achieve six strategic business objectives:

- 1. Operational excellence*
- 2. New products, services, and business models*
- 3. Customer and supplier intimacy*
- 4. Improved decision making*
- 5. Competitive advantage*
- 6. Survival*

What is Business Intelligence?

Operational Excellence



Walmart

- *Power of combining information systems and best business practices to achieve operational efficiency—and over \$524 billion in sales in 2019*
- *Most efficient retail store in world as result of digital links between suppliers and stores*

New Products, Services, and Business Models

- *Information systems and technologies enable firms to create new products, services, and business models*
- *Business model: how a company produces, delivers, and sells its products and services*
- *Example: Apple*
 - *Transformed old model of music distribution with iTunes*
 - *Constant innovations—iPod, iPhone, iPad, etc.*

Customer and Supplier Intimacy

- *Customers who are served well become repeat customers who purchase more*
 - *Example: Mandarin Oriental Hotel*
 - *Uses I T to foster an intimate relationship with its customers, keeping track of preferences, etc.*
- *Close relationships with suppliers result in lower costs*
 - *Examples: Mandarin Oriental Hotel and J C Penney (in text)*
 - *J C Penney uses I T to enhance relationship with supplier in Hong Kong*

Improved Decision Making

Without accurate information, managers must use forecasts, best guesses, and luck, resulting in misallocation of resources, inventory, employees

- *Real-time data improves ability of managers to make decisions*
 - *Example: Verizon's web-based digital dashboard to provide managers with real-time data on customer complaints, network performance, line outages, etc.*

Competitive Advantage

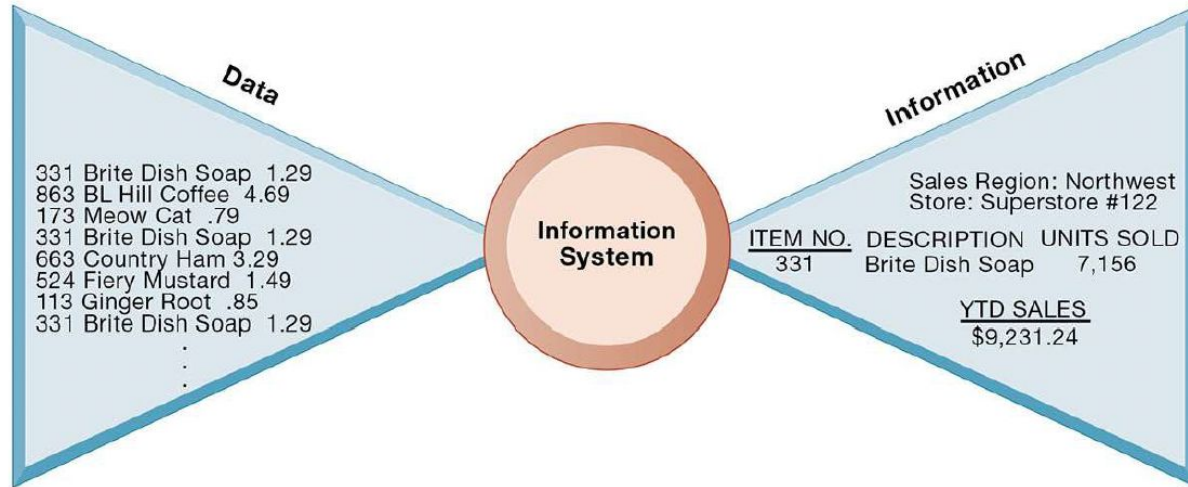
Often results from achieving previous business objectives

- *Advantages over competitors*
 - *Charging less for superior products, better performance, and better response to suppliers and customers*
 - *Examples: Apple, Walmart, U P S are industry leaders because they know how to use information systems for this purpose*

Survival

- *Businesses may need to invest in information systems out of necessity; it is simply the cost of doing business*
- *Keeping up with competitors*
 - *Citibank's introduction of A T Ms*
- *Federal and state regulations and reporting requirements*
 - *Toxic Substances Control Act and the Sarbanes-Oxley Act*

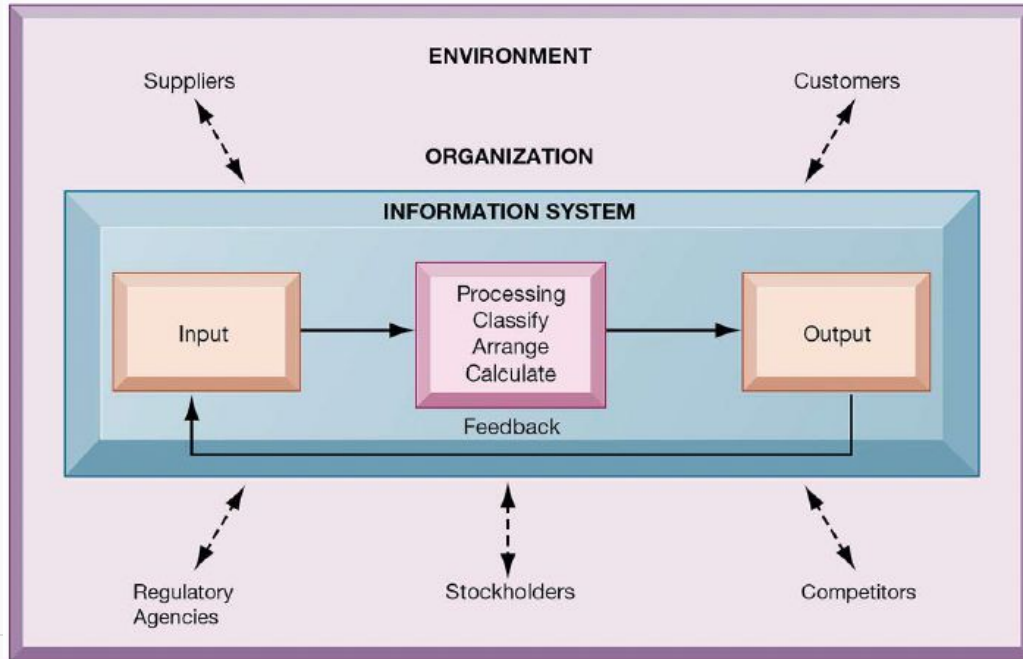
Data and Information



What is Business Intelligence?

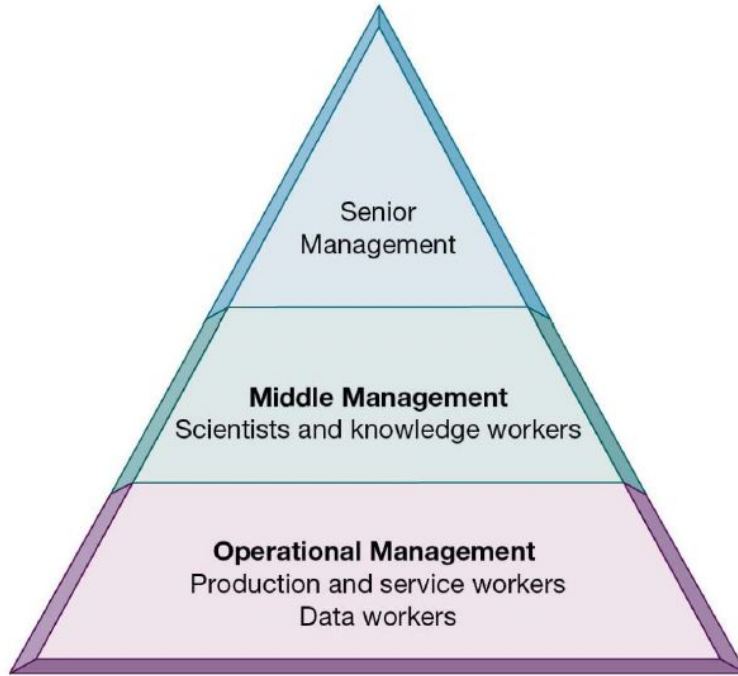
Functions of an Information System

18

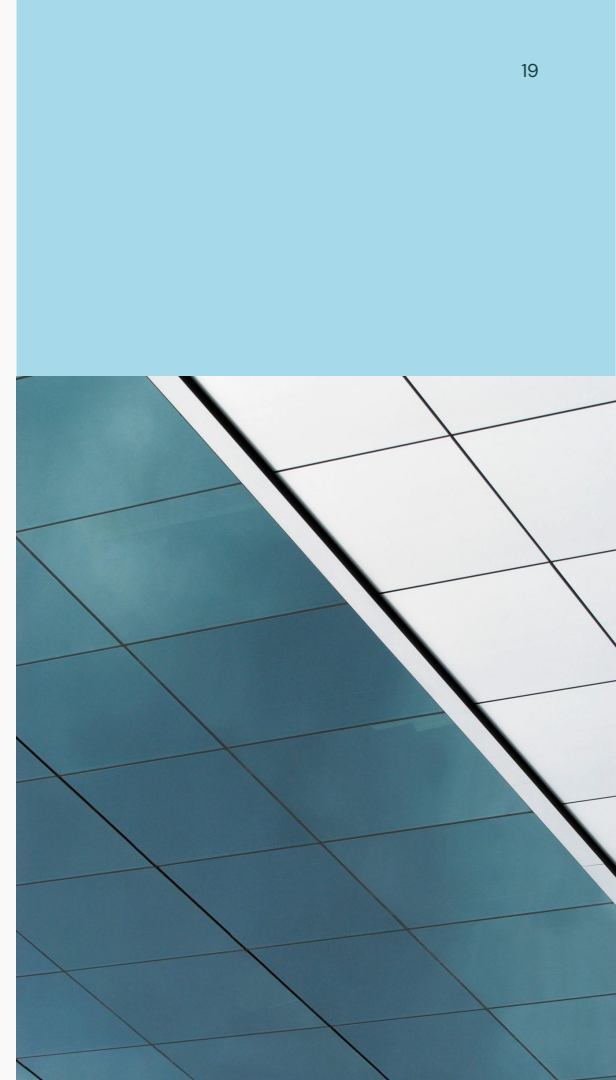


What is Business Intelligence?

Levels in a Firm



What is Business Intelligence?



Management Level	Decision Type	Primary Data Requirements	Characteristics of Data	Typical BI Tools / Outputs	Decisions Made
Senior / Top Management (Executives, CEO, Board)	Strategic Decisions	• Aggregated enterprise-wide data • Historical & trend data • External data (market, competitors, economy)	• Highly summarized • Long-term • Low detail • Often external + internal	• Executive dashboards • Scorecards • KPI summaries • Predictive & prescriptive analytics	• Corporate strategy • Market entry / exit • Mergers & acquisitions • Long-term investment decisions
Middle Management (Department Heads, Managers)	Tactical Decisions	• Department-level data • Periodic performance data • Budget & resource data	• Moderately summarized • Medium-term • Both historical & current	• OLAP reports • Management dashboards • What-if analysis • Diagnostic analytics	• Budget allocation • Performance improvement plans • Sales & marketing campaigns • Resource optimization
Operational Management (Supervisors, Team Leads)	Operational Decisions	• Transaction-level data • Real-time or near real-time data • Process-level data	• Highly detailed • Short-term • Real-time • Mostly internal	• Operational reports • Real-time dashboards • Alerts & notifications	• Daily scheduling • Order processing • Inventory replenishment • Issue resolution

BI Systems vs BI Analytics

Business Intelligence (BI)

“Business intelligence... is descriptive, enabling better business decisions that are based on a foundation of current business data.”

— IBM article comparing BI vs Business Analytics

Business Analytics (BA)

“Business analytics focuses on developing new insights and understanding of business performance... using predictive and prescriptive methods.”

— Wikipedia: Business Analytics

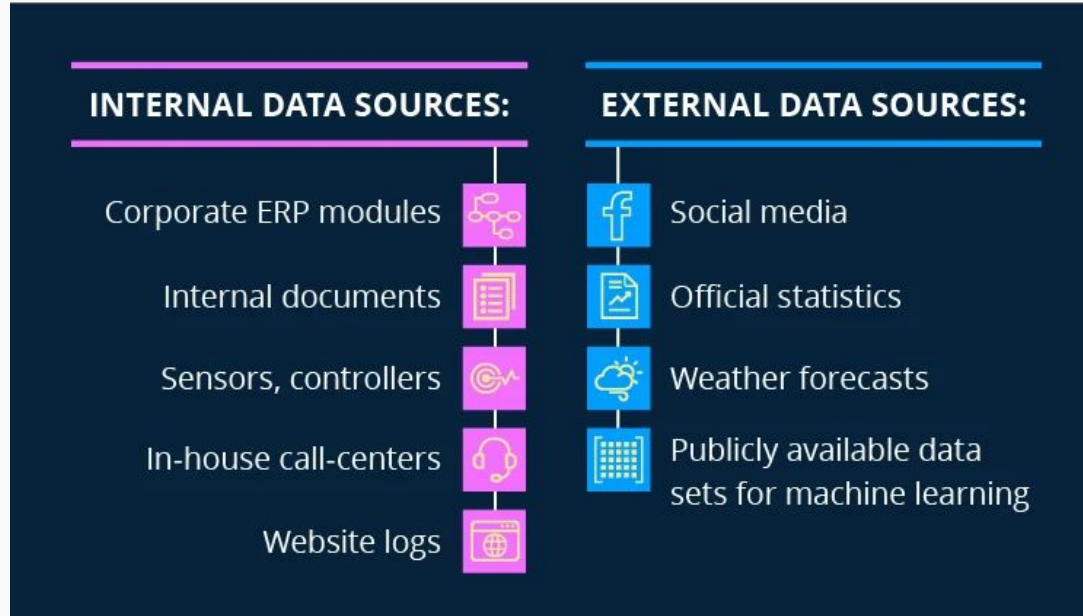
BI Systems vs BI Analytics

According to standard BI descriptions, BI systems include:

- **Data sources** (transaction systems, external data)
- **Data storage** (data warehouses / marts)
- **ETL processes** (to collect, cleanse and integrate data)
- **Analytics engines** (OLAP, data mining, statistical analysis)
- **Presentation layer** (dashboards, reports, visualisations)

BI Systems vs BI Analytics : Data Sources

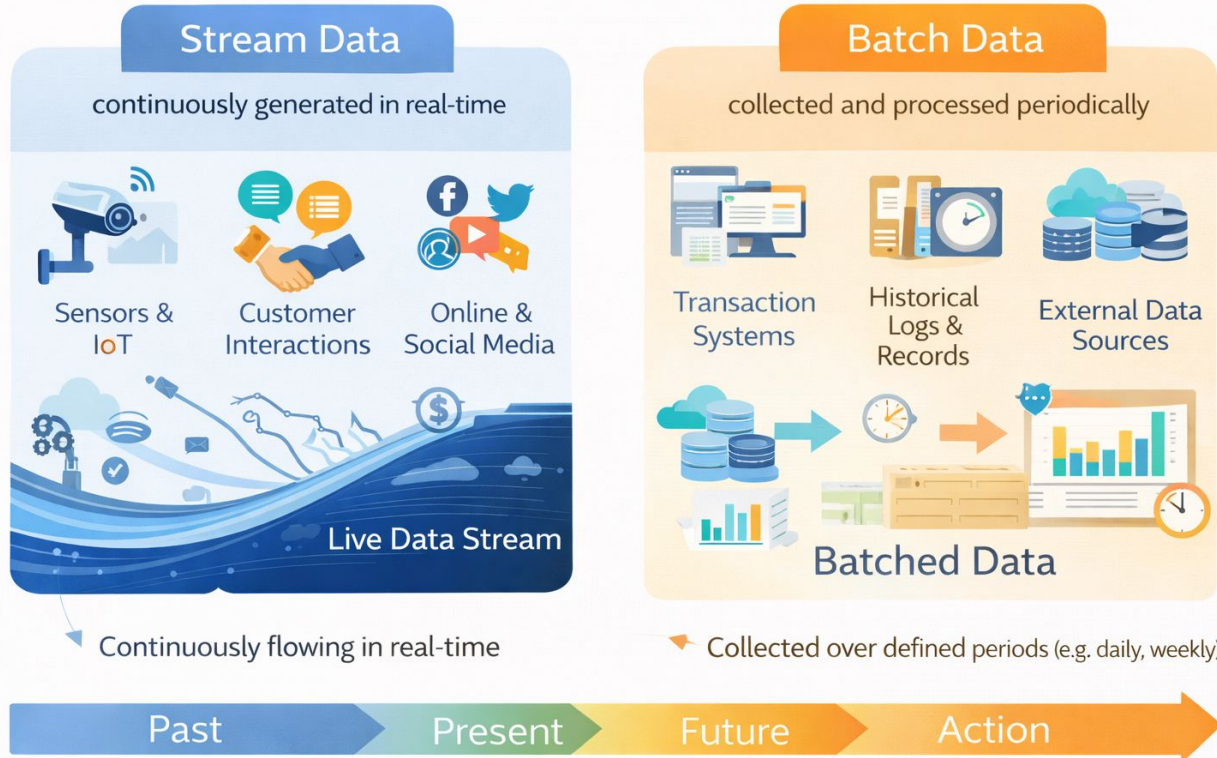
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BI Systems vs BI Analytics : Data Sources

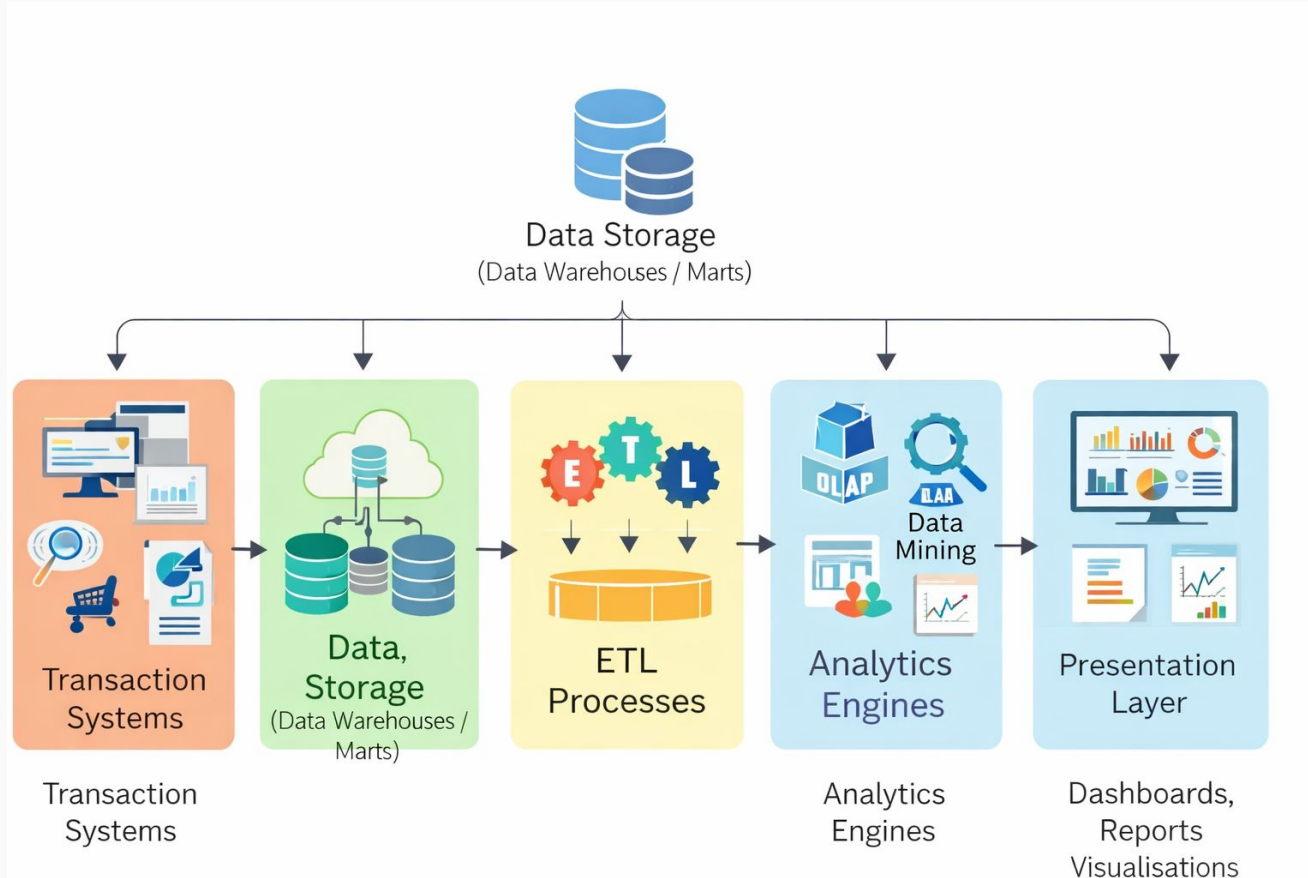
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Different Data Sources



BI Systems vs BI Analytics : BI System

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BI Systems vs BI Analytics : BI Tools

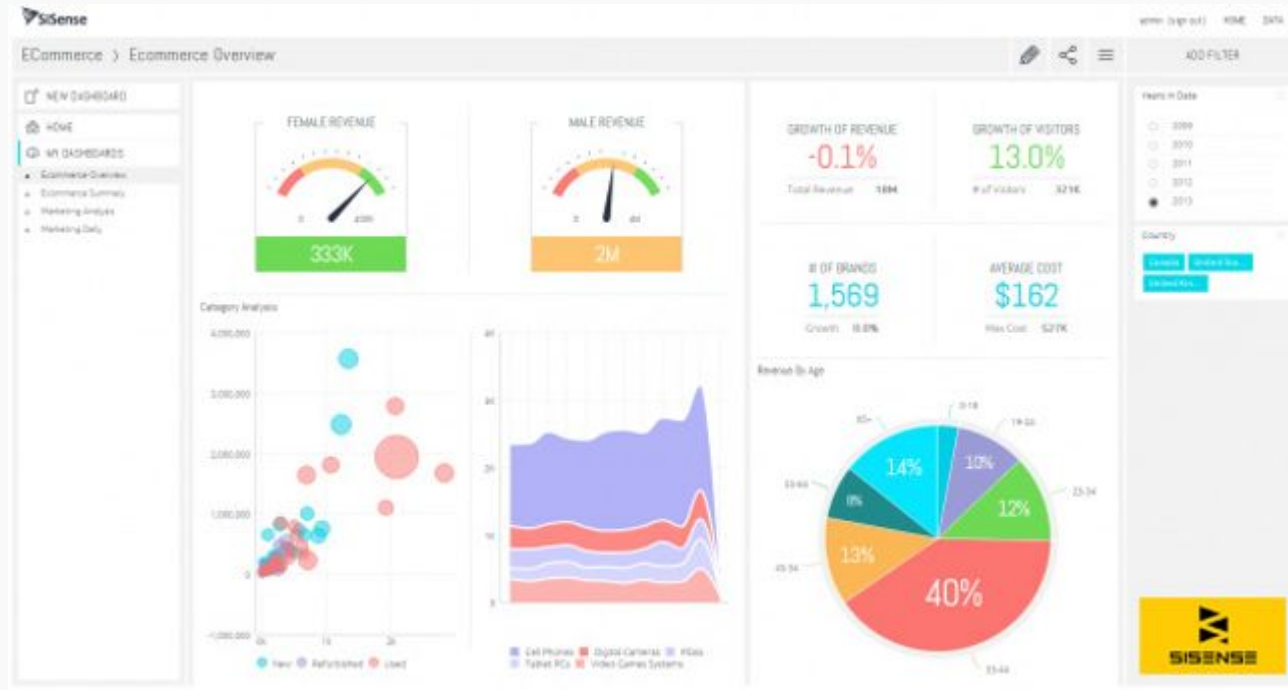
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BI Tools for Data Visualization & Analysis



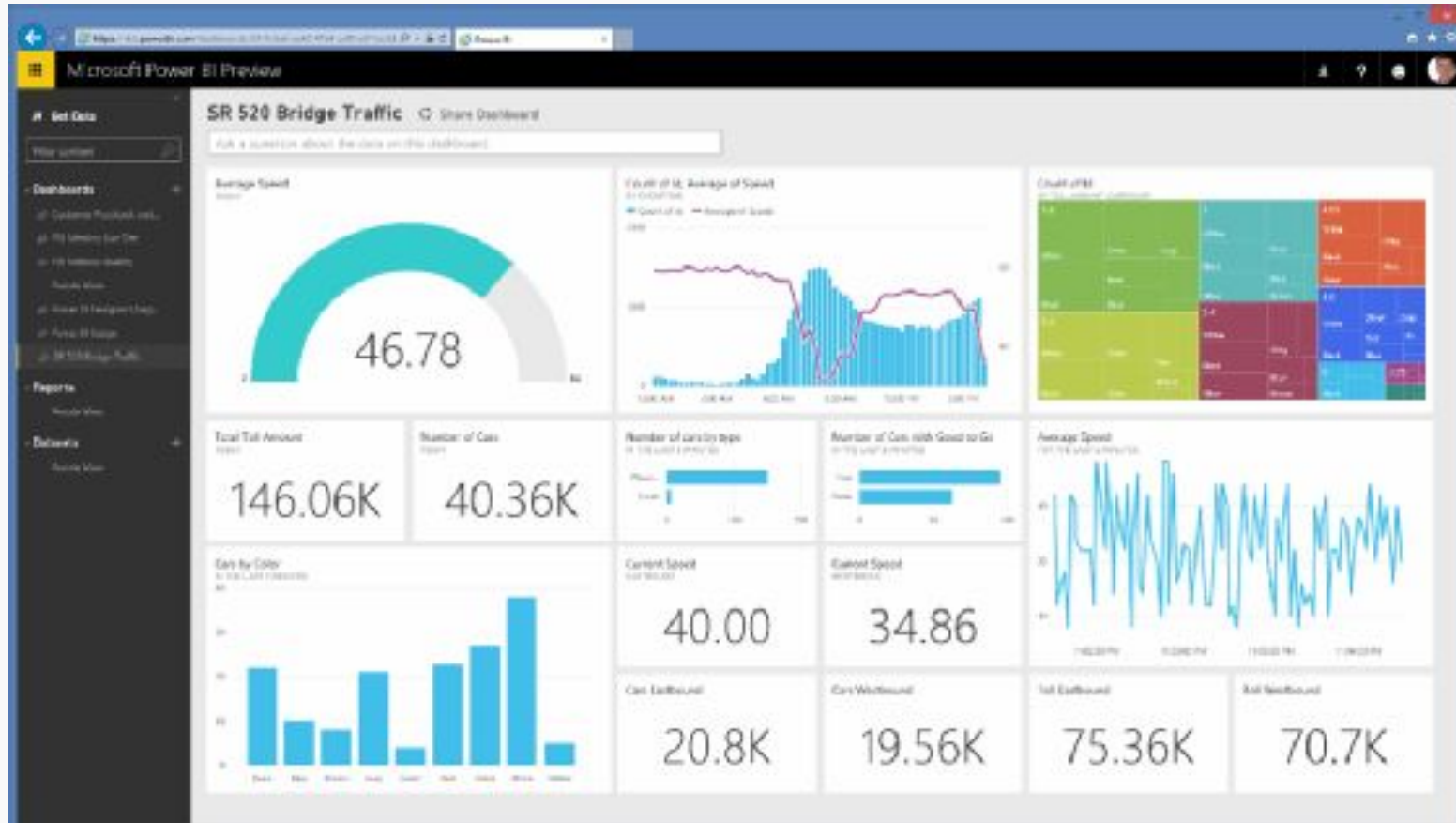
BI Systems vs BI Analytics : BI Tools

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BI Systems vs BI Analytics : BI Tools

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Data Warehouse Tools

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Best Data Warehouse Tools



What is Business Intelligence?

Roles in Business Intelligence

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Data Scientist



Applies statistical and machine learning techniques to analyze and interpret complex data, creates predictive models, and provides actionable insights.

ML Engineer



Designs, builds, and deploys machine learning models, maintains and optimizes ML systems, and ensures the models are scalable and efficient.

Data Engineer



Builds and maintains data pipelines, ensures data is clean and accessible, manages databases, and optimizes data storage and retrieval.

Data Analyst



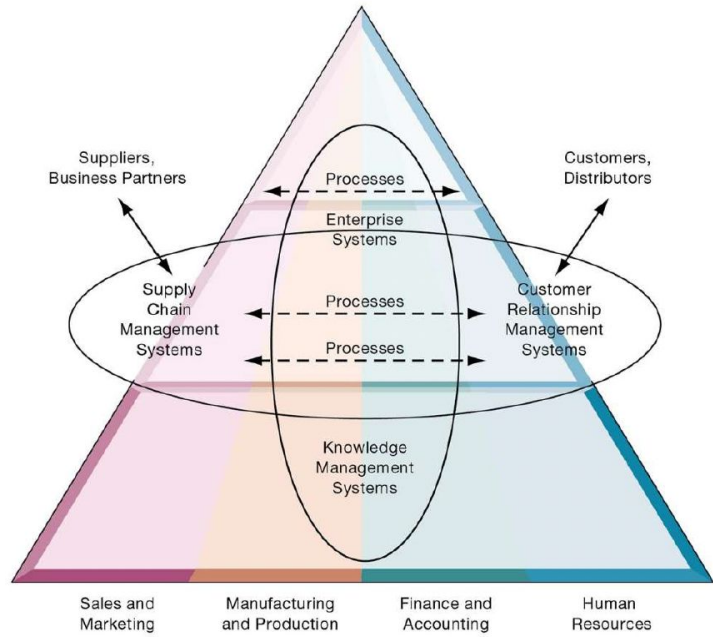
Analyzes and interprets data to identify trends and patterns, creates reports and visualizations, and provides insights to support decision-making.

What is Business Intelligence?

Enterprise Systems

- Also called enterprise resource planning (ERP) systems
- Integrate data from key business processes into single system
- Speed communication of information throughout firm
- Enable greater flexibility in responding to customer requests, greater accuracy in order fulfillment
- Enable managers to assemble overall view of operations

What is Business Intelligence?

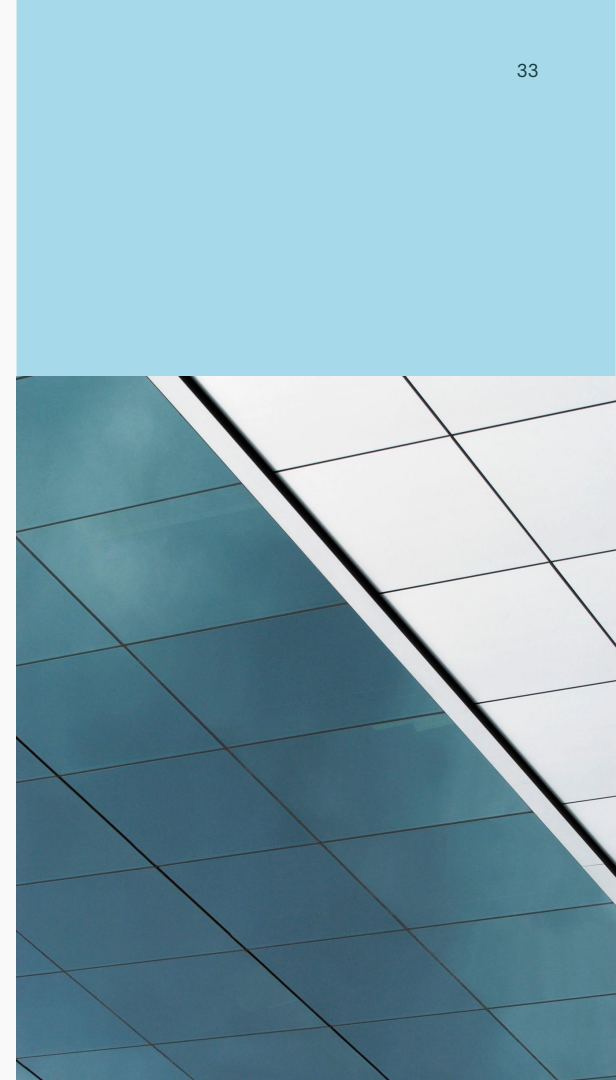


What is Business Intelligence?

Supply Chain Management (SCM) Systems

- Manage relationships with suppliers, purchasing firms, distributors, and logistics companies
- Manage shared information about orders, production, inventory levels, and so on
- Goal is to move correct amount of product from source to point of consumption as quickly as possible and at lowest cost
- Type of interorganizational system: Automating flow of information across organizational boundaries

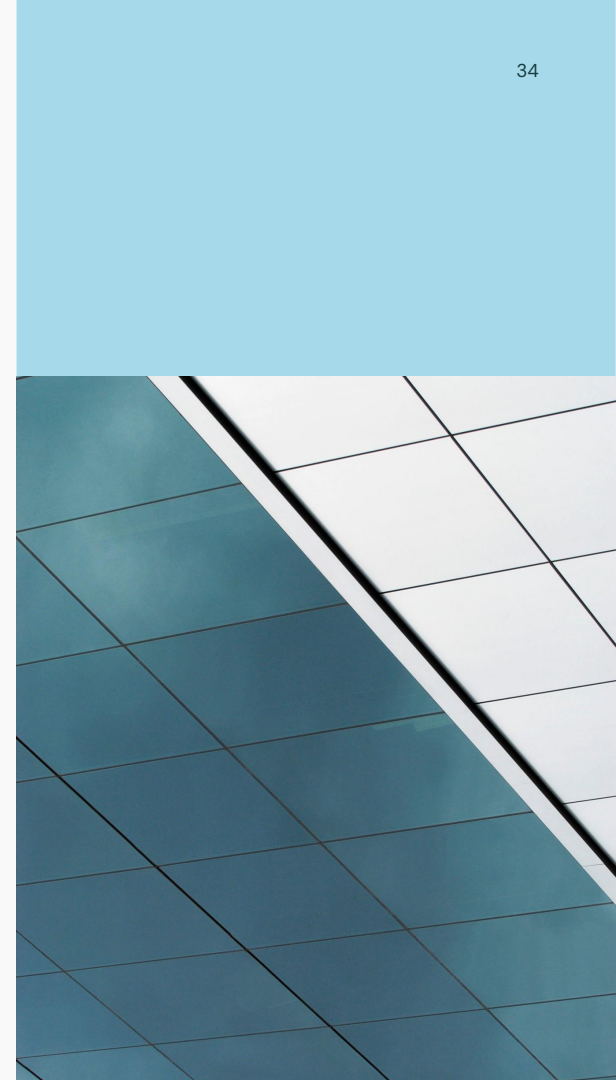
What is Business Intelligence?



Customer Relationship Management (CRM) Systems

- Help manage relationship with customers
- Coordinate business processes that deal with customers in sales, marketing, and customer service
- Goals:
 - Optimize revenue
 - Improve customer satisfaction
 - Increase customer retention
 - Identify and retain most profitable customers
 - Increase sales

What is Business Intelligence?



Knowledge Management Systems (KMS)

- Manage processes for capturing and applying knowledge and expertise
- Collect relevant knowledge and make it available wherever needed in the enterprise to improve business processes and management decisions
- Link firm to external sources of knowledge

What is Business Intelligence?

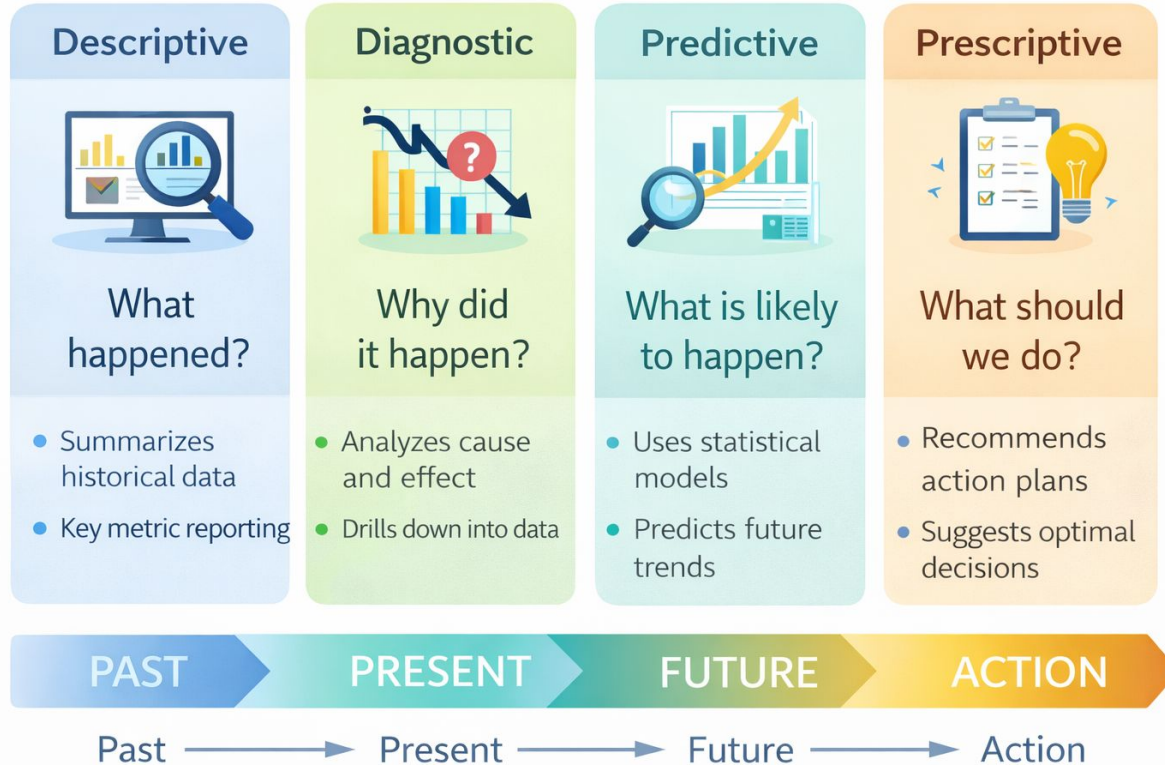
Business Value

Driving Business Value with Modern BI Tools: A Real-World Example



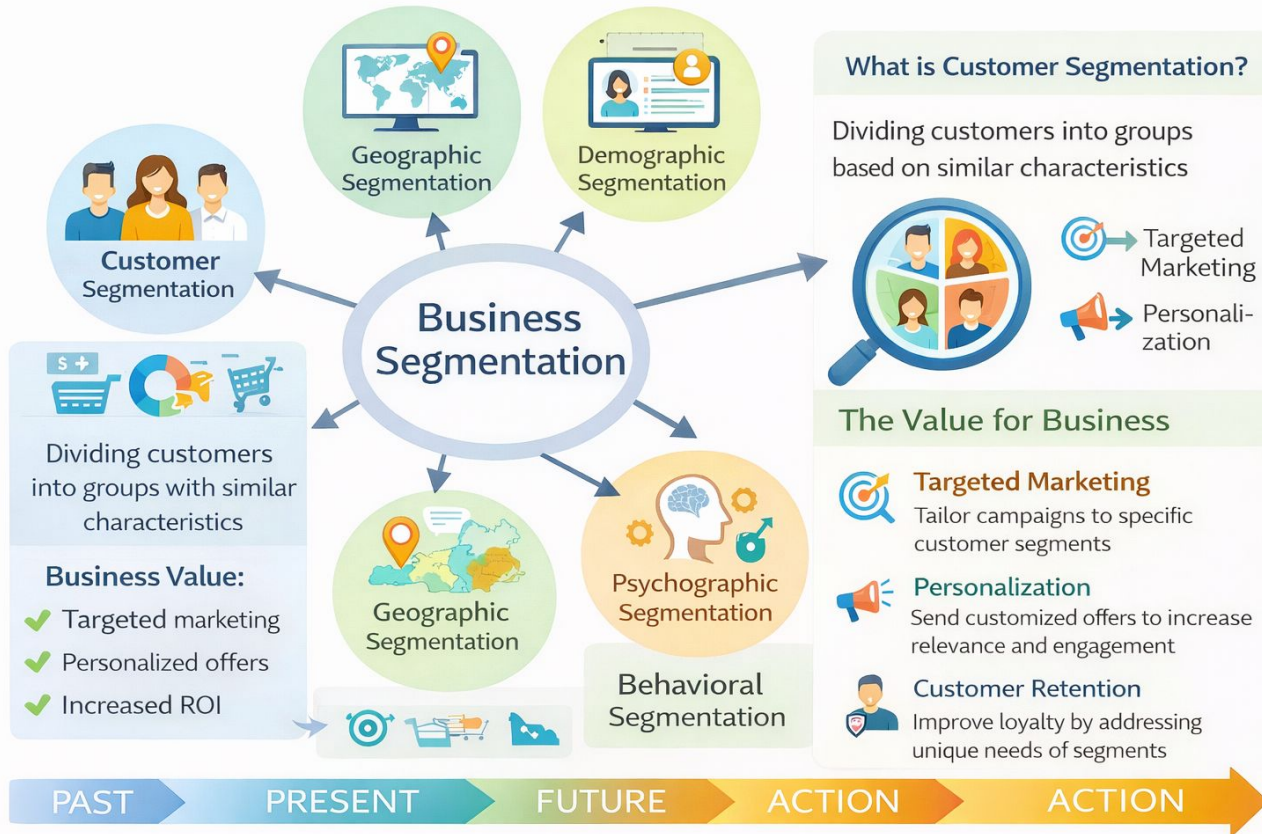
Business Value : Analytics

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Business Value : Segmentations

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Case Study 1



Amazon – Dynamic Pricing & Real-Time Analytics

Amazon uses **real-time data** and analytics to adjust prices across millions of products based on supply, demand, and competitor pricing.

[Read how Amazon does this here: *How Amazon Uses Real-Time Data And Dynamic Pricing...*](#)



This demonstrates BI + operational analytics in action — dynamic pricing decisions are delivered rapidly to optimize revenue.

Case Study 2



Uber – BI & Real-Time Analytics for Operations

Uber continuously processes **real-time data** from riders and drivers for:

- Route optimization
- Dynamic (surge) pricing
- ETA and matching algorithms

[Read about the real-time analytics behind this here: *How Uber uses real-time analytics for dynamic pricing...*](#)



This shows how operational BI supports **instant decision execution** in a distributed platform.

Thank you